L/T/LT/TT-Separation Analysis Procedure

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March 6, 2026

1 Introduction

A central challenge of modern physics involves understanding the fundamental building blocks of matter (protons and neutrons) in our universe. It is well understood that visible matter (atoms) is composed of different combinations of elementary particles (electrons (e), quarks (q), and gluons (g)) as described by the Standard Model. In the Standard Model, quarks and gluons interact with each other through the strong nuclear force, the theory which describes the strong force and its interaction is known as Quantum Chromodynamics (QCD). While QCD is an extremely successful theory, it struggles to predict the interactions of quarks and gluons at long distances (low energies). The pion (π) offers an excellent opportunity to gain deeper insight into QCD in low-energy regime. It is a relatively simple hadron and has an important role in nature, particularly in understanding the forces that bind nuclear matter together. The pion form factor (F_π) is an important observable to understand the inner structure of pions. To measure F_π at high four-momentum transfer Q^2, we performed the Pion-LT experiment (E12-19-006) in Hall C of Jefferson Lab. This experiment is the subject of my thesis. This experiment seeks to extract F_π in a previously inaccessible region of four-momentum transfer (Q^2). Due to its scientific significance, this experiment was characterized as "early high impact" by the JLab PAC and thus is a high-priority experiment. The Pion-LT experiment collected data over a broad range of Q^2 values, from $0.38\,\mathrm{GeV}^2$ up to $8.50\,\mathrm{GeV}^2$.

2 The Pion-LT Experiment

The Pion-LT experiment (E12-19-006) was designed to measure F_π from $Q^2 = 1.45\,\mathrm{GeV}^2$ to high $Q^2 = 8.50\,\mathrm{GeV}^2$. This experiment is one of the most important experiments carried out at Experimental Hall C in Jefferson Lab. F_π is extracted from the exclusive pion (π^+) electro-production reaction $p(e, e', \pi^+)n$. In the reaction $p(e, e'\pi^+)n$, an incident electron (e) emits a virtual photon, which interacts with the target proton, resulting in a scattered electron (e'), a positively charged pion (π^+), and a neutron (n). The electrons are detected in the HMS spectrometer while the pions are detected in the SHMS spectrometer as shown in Figure 1.

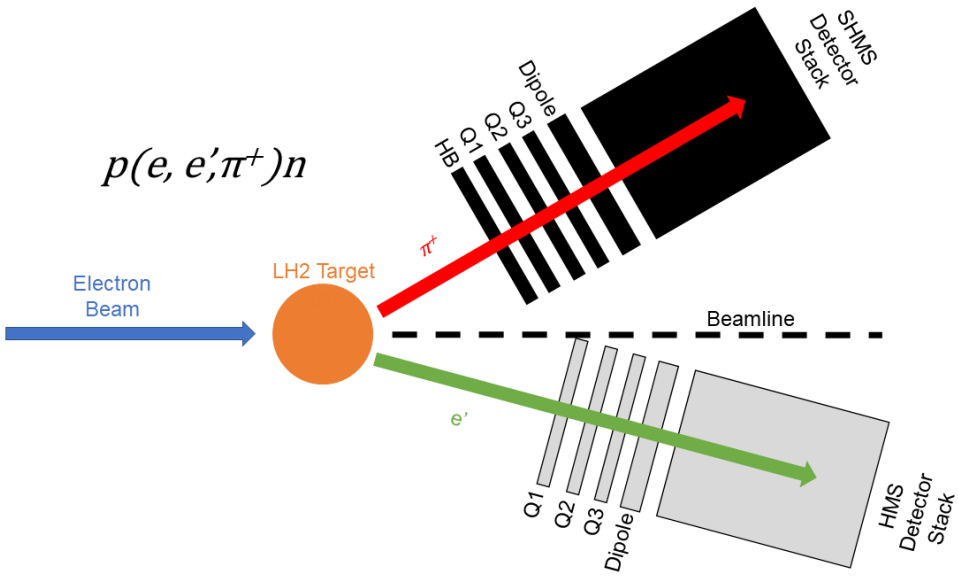


Figure 1: The exclusive pion (π^+) electro-production reaction in Hall C.

The neutrons are not directly detected; they are studied in Hall C by reconstructing the missing mass of the reaction with sufficient resolution to exclude any competing reaction. The missing mass equation is given as follows:

$$M_{miss} = \sqrt{(E_{beam} + m_p - E_{e'} - E_{\pi^+})^2 - (\overrightarrow{p}_e - \overrightarrow{p}_{e'} - \overrightarrow{p}_{\pi^+})^2} \tag{1}$$

A selection cut around the neutron missing mass peak will be applied to isolate a clean sample of exclusive pion events. The total physical cross-section for the exclusive pion electro-production reaction is expressed using the Rosenbluth formula as follows:

$$2\pi \frac{d^2\sigma}{dtd\phi} = \epsilon \frac{d\sigma_L}{dt} + \frac{d\sigma_T}{dt} + \sqrt{2\epsilon(\epsilon+1)}\frac{d\sigma_{LT}}{dt}\cos\phi + \epsilon\frac{d\sigma_{TT}}{dt}\cos 2\phi, \tag{2}$$

$$\text{where} \quad \epsilon = 1 + 2\frac{(E_e - E_{e'})^2 + Q^2}{Q^2}\tan^2\frac{\theta_{e'}}{2} \tag{3}$$

To perform a Rosenbluth L/T separation, experimental data must be acquired at a minimum of two different beam energies and at various angles relative to the virtual photon's direction of propagation. Achieving a reliable L/T/LT/TT separation requires complete knowledge of the reaction kinematics, particularly the virtual photon polarization (ϵ), Q^2, W, t and the azimuthal angle (ϕ) of the pion. In Eqn 2, σ_L and σ_T represent the longitudinal and transverse components of the cross-section, while σ_{LT} and σ_{TT} are interference terms. Only σ_L is directly sensitive to F_π. To extract F_π from the data, one has to do a precise L/T separation to determine σ_L, and then use a theoretical model to extract the pion form factor from data. This indirect method is the only practical approach to measure F_π at high Q^2. The results from the Pion-LT experiment will rigorously test existing theoretical models and help to develop future improved hadronic structure models.

3 L/T/LT/TT Separation Analysis Overview

Before the L/T/LT/TT separation, the ratio method (Eqn4) is used to perform all of the spectrometer acceptance, target-length, and bin-containment corrections to the data. This method is reliable provided that the simulation accurately represents the data. An iterative procedure is used to produce a simulation that accurately represents the data, where the discrepancy between data and simulation decreases with each iteration.

$$\sigma_{exp} = \frac{Y_{exp}}{Y_{sim}} \sigma_{sim} \tag{4}$$

The L/T/LT/TT separation of the total differential cross-section is an iterative process that requires gradual improvement of the estimated cross-section parameterization in the simulation by comparison with the data. The resulting improved simulation should provide an acceptable description of the experimental data across the full spectrometer acceptance and kinematic coverage. The iterative procedure is summarized in the flow chart shown in the Figure 2.

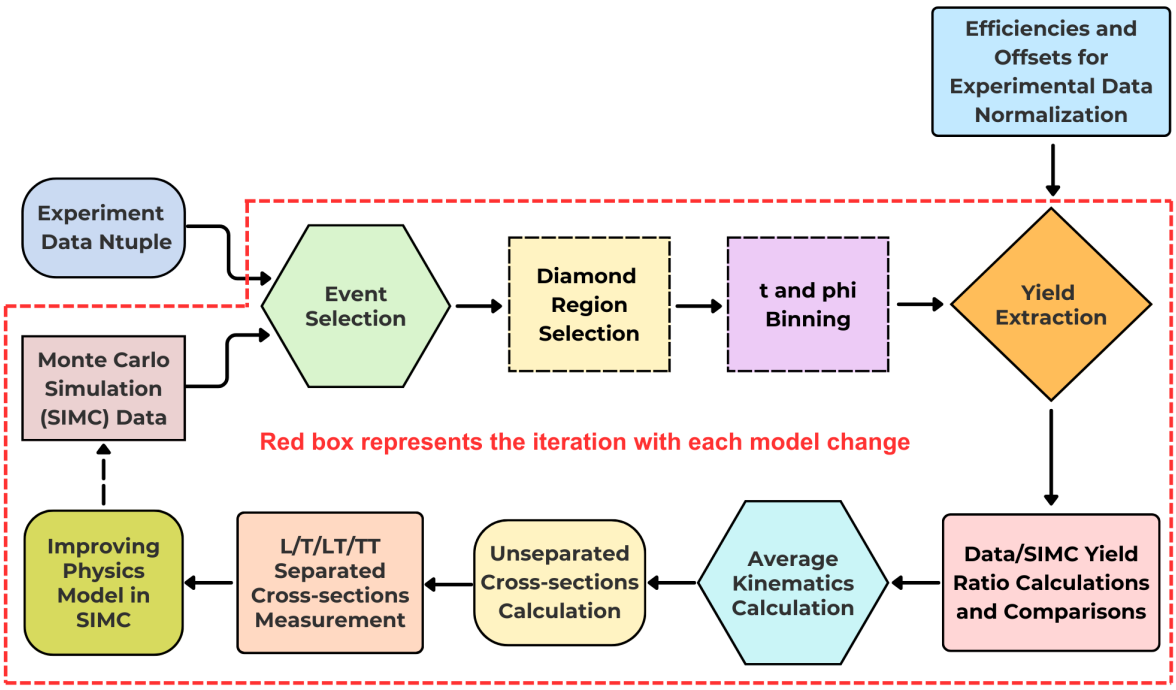
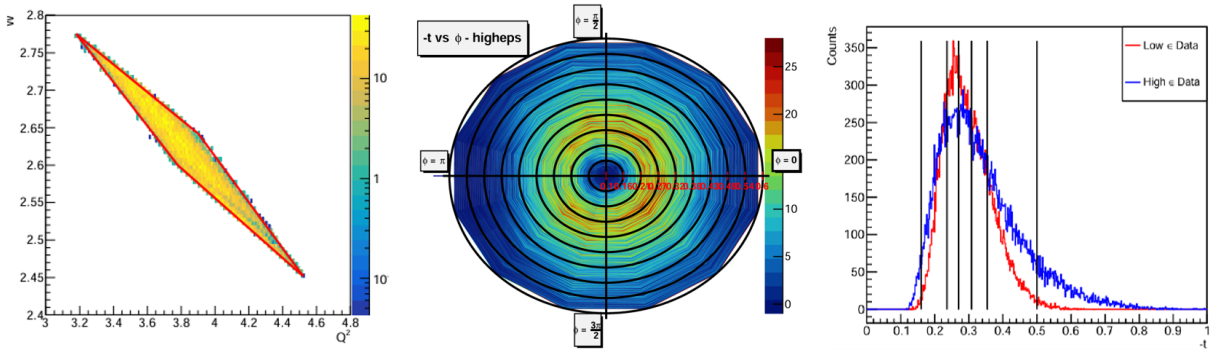


Figure 2: Flowchart of the iterative L/T/LT/TT separation procedure. The red-highlighted box indicates the steps repeated for each updated model in the Monte Carlo simulation (SIcrossMC).

Before the L/T/LT/TT separation, the first step is to perform detector calibrations of the collected experimental data, including the Drift Chambers, Cherenkov detectors, Hodoscopes, and Calorimeters. The L/T/LT/TT separation analysis is very challenging and requires careful attention to systematic studies. In the second step, systematic studies are carried out, including spectrometer acceptance, kinematics, efficiencies, offset calculation, particle identification (PID), elastic $H(e, e'p)$ reaction studies, etc.

At the start of the L/T/LT/TT separation procedure, the experimental and simulated ntuples are generated using the HallC analyzer (HCANA) and the Monte Carlo simulation



(a) (b) (c)

Figure 3: (a) Q^2 vs W distribution. Red lines indicate the diamond cut, applied to equalize the experimental acceptance for both low-ϵ and high-ϵ settings. (b) Left, right, and centre combined t vs ϕ plot from high-ϵ setting. (c) t distribution plot. The red and blue distributions correspond to the low-ϵ and high-ϵ settings, respectively. The black vertical line shows five $t$bins. All these plots are from "$Q^2 = 3.85, W = 2.62, t = 0.21$" physics setting.

(SIMC) for the required physics kinematic settings. The next step is to apply PID cuts for event selection. In this step, appropriate cuts are applied to select electrons (e^-) in the HMS spectrometer in coincidence with pions (π^+) in the SHMS spectrometer, in order to obtain a clean $e\pi$ event sample.

To perform a Rosenbluth L/T separation, experimental data must be acquired at a minimum of two different beam energies and at various angles relative to the virtual photon's direction of propagation. Most of the physics data were taken on three different SHMS angles: left, center and right to get the full ϕ coverage by keeping in mind the angle restrictions of the SHMS as shown in Figure 3(b). However, the accessible phase-space is constrained by the spectrometer acceptances, which vary with beam energy. Higher spectrometer momenta result in the broad ΔP acceptances compared to lower momenta. To ensure consistency across different kinematic settings and to reduce systematic uncertainties arising from variations in W coverage, a "diamond cut" is applied in the next step. This cut restricts the analysis to the overlapping phase-space region common to all relevant ϵ settings. The diamond-like selection region is illustrated in Figure 3(a), showing the events that satisfy this criterion. The same diamond cut is applied to SIMC. Only events within these overlapping regions are used to extract yields and separated cross-sections. It is important to confirm that the SIMC and data diamond acceptances match. If there is any discrepancy, the smaller of the two regions must be used. This ensures consistent experiment acceptances across different beam energies and improves the reliability of the L/T separation.

The next step is to bin the experimental data and SIMC in t and ϕ. As an example, the "$Q^2 = 3.85, W = 2.62, t = 0.21$ (2-ϵ)" setting is binned into five t-bins and 15 ϕ-bins as shown in Figure 3(b) and (c). This binning scheme is determined by the available experimental statistics. Fine binning in t also depends on the t-resolution, which is evaluated from the SIMC by calculating the standard deviation (σ) of the difference between the measured t and the initial t_i values generated in the simulation. Figure 3(c) illustrates the final t-binning applied to two ϵ-settings of "$Q^2 = 3.85, W = 2.62, t =$

0.21", where the red and blue distributions correspond to the low-ϵ and high-ϵ settings, respectively. The binning in t is designed such that each bin contains approximately equal statistics (>2500 events), while also ensuring that the bin widths are not smaller than the t-resolution determined from SIMC. This approach maintains both statistical consistency and physical reliability in the extraction of the separated cross-sections. The next step is to further subdivide each t-bin into ϕ-bins for the extraction of the yields and cross-sections and to proceed with the remaining steps of the analysis.

The next step is to calculate the normalized yields for each bin of experimental data and SIMC. The normalized yield is defined as:

$$Y_{exp} = \frac{N}{Q_{eff}}, \quad Q_{eff} = Q_{tot} \cdot \epsilon_{tot} \tag{5}$$

where N is the total number of normalized events that successfully passes the event selection criteria, Q_{eff} is the effective charge, Q_{tot} is the cumulative charge accumulated at the target and ϵ_{tot} is the overall efficiency. The experimental charge normalized yield must be corrected for the tracking and detector efficiencies in order to compare it to the SIMC yield. Next, the yield ratios are calculated, and comparison plots are generated for various kinematic variables to verify that the data and SIMC are consistent. The bin-by-bin average kinematic quantities Q^2, t, θ, ϕ, the DATA/SIMC ratios, etc., are used to calculate the total differential cross section for each t- and ϕ-bin. Next, the Rosenbluth equation is fitted to these unseparated cross-sections to extract the L/T/LT/TT separated cross-sections. The SIMC model functional forms for each separated cross-section (σ_L, σ_T, σ_{LT}, and σ_{TT}) are then fitted in ROOT to obtain new optimized parameters for the next iteration. Using these parameters, new SIMC unseparated cross-sections are calculated, the SIMC events are reweighted accordingly, and the procedure is repeated iteratively. At each iteration, the DATA/SIMC yield ratios and comparison plots of spectrometer and kinematics quantities must be examined to check the validity of the model. If, after several iterations, the model still does not adequately describe the separated cross-sections, the physics model in SIMC must be modified empirically and the entire procedure repeated.

4 L/T/LT/TT Separation Analysis Framework

Before starting the analysis, a virtual environment must be set up on the JLab computing account (ifarm). All the required steps can be found at `https://redmine.jlab.org/projects/hall-c/wiki/Alma9_Setup_on_ifarm`. There are four main GitHub repositories required to start the L/T/LT/TT separation analysis:

hallc_replay_lt	–	`https://github.com/JeffersonLab/hallc_replay_lt`
UTIL_PION	–	`https://github.com/JeffersonLab/UTIL_PION`
UTIL_BATCH	–	`https://github.com/JeffersonLab/UTIL_BATCH`
LTSEP_ANALYSIS	–	`https://github.com/junaid-muhammad/LTSEP_ANALYSIS`
simc_gfortran	–	`https://github.com/JeffersonLab/simc_gfortran`

The main repository is `hallc_replay_lt`, and the `UTIL_PION`, `UTIL_BATCH` and `LTSEP_ANALYSIS` repositories should be cloned inside this directory. The `simc_gfortran` repository can

be set up anywhere on the ifarm. The `hallc_replay_lt` repository is primarily used to perform physics replays, converting the raw data (.dat) files into analyzable ROOT format. The details about the Monte Carlo `simc_gfortran` setup and usage can be found at `https://hallcweb.jlab.org/wiki/index.php/SIMC_Monte_Carlo`.

The **UTIL_BATCH** repository contains all the PionLT runlists, located at path: `UTIL_BATCH/InputRunLists/PionLT_2021_2022/`. It is recommended to keep the runlist naming convention, for example `Q3p85_W2p62_t0p21_loweps_center` for physics data and `Q3p85_W2p62_t0p21_loweps_center_dummy` for dummy data, so that the shell scripts can construct the required filenames from this argument. The Overview of L/T/LT/TT separation analysis framework is illustrated in Figure 4.

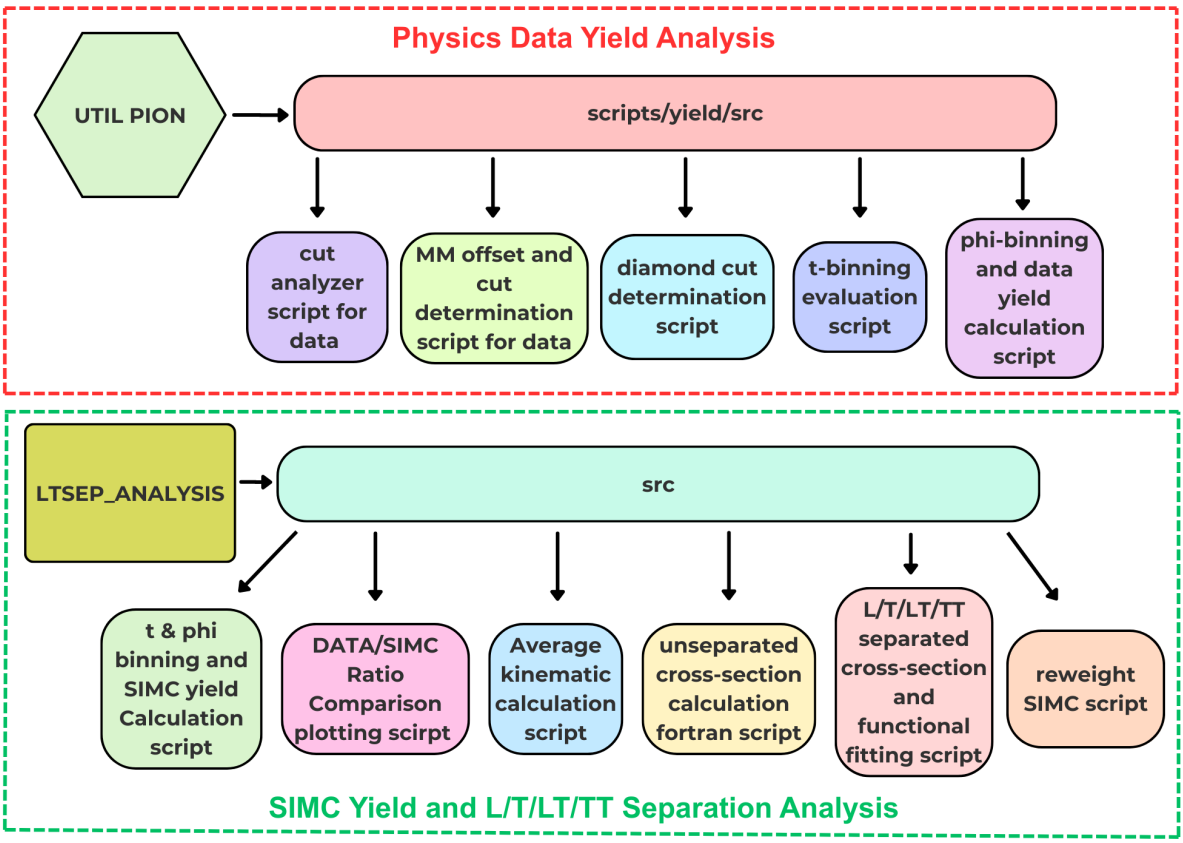


Figure 4: Overview of L/T/LT/TT separation analysis framework.

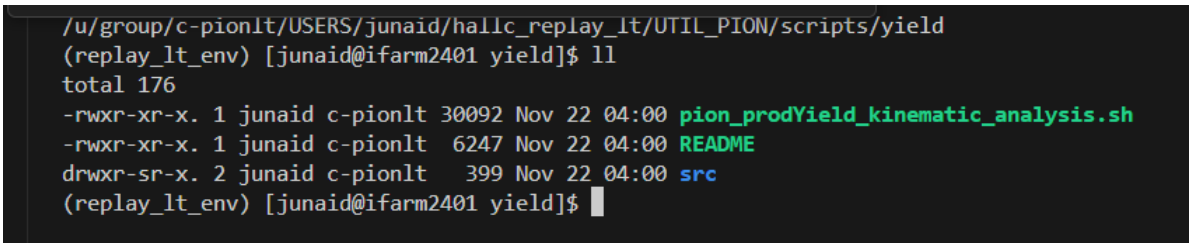
4.1 Physics Data Yield Analysis

In this subsection, the normalized data yield analysis in the L/T/LT/TT separation will be discussed. In the `UTIL_PION` repository, there are two important directories: **efficiencies** and **LTSep_CSVs**. The `efficiencies` folder contains all the calculated efficiencies required to determine the normalized data yields. The `LTSep_CSVs` directory stores all the CSV files, generated by the analysis scripts, that are used in the L/T/LT/TT separation analysis for the required physics settings. All analysis scripts required to calculate the binned normalized yields are located in the **scripts/yield** directory.

The `yield` directory contains a subdirectory **src** and the main analysis shell script **pion_prodYield_kinematic_analysis.sh** (Figure 5). The shell script has nine flags,

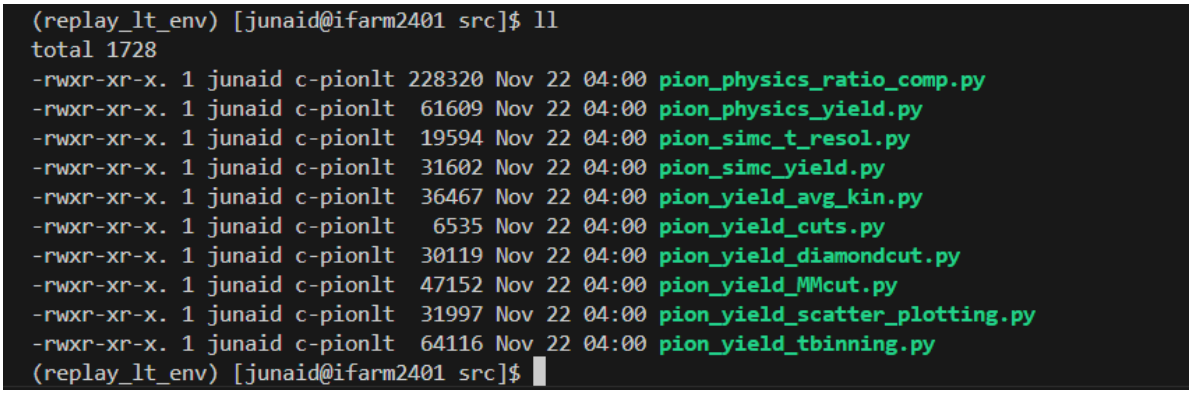
and each flag calls a separate Python script in the `src` subdirectory, depending on the analysis requirements. All the required paths, file-name constructions, and Python script arguments are declared in this shell script. The details of the flags are explained in the Table 1. The script requires three arguments. To run this script, execute the following command:

```
./pion_prodYield_kinematic_analysis.sh -flag <RunType(Prod)> <RunList>
```



```
/u/group/c-pionlt/USERS/junaid/hallc_replay_lt/UTIL_PION/scripts/yield
(replay_lt_env) [junaid@ifarm2401 yield]$ ll
total 176
-rwxr-xr-x. 1 junaid c-pionlt 30092 Nov 22 04:00 pion_prodYield_kinematic_analysis.sh
-rwxr-xr-x. 1 junaid c-pionlt  6247 Nov 22 04:00 README
drwxr-sr-x. 2 junaid c-pionlt   399 Nov 22 04:00 src
(replay_lt_env) [junaid@ifarm2401 yield]$
```

Figure 5: screenshot of `UTIL_PION/scripts/yield`



```
(replay_lt_env) [junaid@ifarm2401 src]$ ll
total 1728
-rwxr-xr-x. 1 junaid c-pionlt 228320 Nov 22 04:00 pion_physics_ratio_comp.py
-rwxr-xr-x. 1 junaid c-pionlt  61609 Nov 22 04:00 pion_physics_yield.py
-rwxr-xr-x. 1 junaid c-pionlt  19594 Nov 22 04:00 pion_simc_t_resol.py
-rwxr-xr-x. 1 junaid c-pionlt  31602 Nov 22 04:00 pion_simc_yield.py
-rwxr-xr-x. 1 junaid c-pionlt  36467 Nov 22 04:00 pion_yield_avg_kin.py
-rwxr-xr-x. 1 junaid c-pionlt   6535 Nov 22 04:00 pion_yield_cuts.py
-rwxr-xr-x. 1 junaid c-pionlt  30119 Nov 22 04:00 pion_yield_diamondcut.py
-rwxr-xr-x. 1 junaid c-pionlt  47152 Nov 22 04:00 pion_yield_MMcut.py
-rwxr-xr-x. 1 junaid c-pionlt  31997 Nov 22 04:00 pion_yield_scatter_plotting.py
-rwxr-xr-x. 1 junaid c-pionlt  64116 Nov 22 04:00 pion_yield_tbinning.py
(replay_lt_env) [junaid@ifarm2401 src]$
```

Figure 6: screenshot of `UTIL_PION/scripts/yield/src`

The subdirectory `src` contains 9 main Python scripts (Figure 6) required by the **pion_prodYield_kinematic_analysis.sh** shell script. The **pion_yield_cuts.py** script trims the replayed ROOT files by applying various cuts, including acceptance, coincidence time, HMS detector (Cherenkov and Calorimeter), SHMS detector (Aerogel), and RF time cuts. The **pion_yield_MMcut.py** script compares the data missing-mass distribution with the SIMC missing-mass distribution in order to determine the missing-mass offset and cut. These scripts must be run separately for each physics setting (left, center and right). The **pion_yield_diamondcut.py** script determines the vertices for the diamond cut. It provides an initial estimate of the vertices, which must then be adjusted manually in the script by looking at the output plots. The **pion_simc_t_resol.py** script checks the t-resolution (t- t_0) for SIMC. The **pion_yield_tbinning.py** script determines the t-binning for a given physics setting; the bin widths must be adjusted manually in the script. Each t bin should have a similar number of events and should be greater than 2500. The **pion_physics_yield.py** script performs the ϕ-binning and calculates the normalized physics yields for each t- and ϕ-bin. The **pion_simc_yield.py** script calculates the SIMC yields for each t- and ϕ-bin for the default SIMC physics

Flag	Function	RunList
-c	runs cut analyser script to trim down the ROOT file for further analysis	Setting by setting
-m	runs missing-mass check script to determine the cut and offset for the missing-mass distribution	Setting by setting
-d	runs diamond-cut check script to determine the diamond cut for data and SIMC	Setting by setting
-e	runs t-resolution script to determine the t-resolution $(t - t_0)$ for SIMC	Whole setting
-t	runs t-binning script to determine the t-binning for data	Whole setting
-p	runs ϕ-binning and physics-yield script to determine the normalized physics yields of data	Setting by setting
-s	runs SIMC-yield script to determine SIMC yields for the default SIMC physics model (`physics_pion.f`)	Setting by setting
-r	runs plotting script to compare DATA/SIMC and determine ratios for the default SIMC physics model	Setting by setting
-a	runs script to calculate average kinematics and yields for LTSep analysis for the default SIMC physics model	Whole setting

Table 1: Description of the flags used in `pion_prodYield_kinematic_analysis.sh`. Note: "setting by setting" means running separately for each left, right, and center setting for a given Q^2 and W. "Whole setting" refers to the complete Q^2, W kinematic setting (no need to specify left, center, and right) and should be run only once.

model (`physics_pion.f`). The **pion_physics_ratio_comp.py** script calculates the DATA/SIMC yield ratios and produces comparison plots for the default SIMC physics model. The **pion_yield_avg_kin.py** script computes the average kinematics for each bin and stores them in the required format for the subsequent unseparated cross-section calculation by Fortran scripts and L/T/LT/TT separation analysis. The bin-by-bin physics data yields need to be calculated only once. During the iterations, the bin-by-bin SIMC yields will change when the model parameterization is updated or the functional forms of the separated cross-sections are modified. This part will be discussed in the next section.

Note: Each script typically produces three types of output files: ROOT files, PDF plots, and a CSV file. The ROOT files and PDFs are stored at path:/volatile/hallc/c-pionlt/$USER/OUTPUT/Analysis/PionLT, and the CSV files are written to the path: /u/group/c-pionlt/USERS/$USER/hallc_replay_lt/UTIL_PION/LTSep_CSVs.

4.2 SIMC Yields and L/T Separated Cross-section Analysis

In this subsection, the iteration in the L/T/LT/TT separation analysis will be discussed. In the `LTSEP_ANALYSIS` repository, there are 3 important subdirectories: **iterations**, **LTSep_CSVs**, and **src**, as well as the main **pion_prodYield_ltsep_analysis.sh** shell script (Figure 7).

```
total 240
drwxrwsr-x.  3 junaid c-pionlt    46 Nov 22 03:48 iterations
drwxrwsr-x. 11 junaid c-pionlt   308 Nov 22 03:32 LTSep_CSVs
-rwxrwxr-x.  1 junaid c-pionlt 27656 Nov 22 01:51 pion_prodYield_ltsep_analysis.sh
-rw-rw-r--.  1 junaid c-pionlt    14 Oct 26 21:34 README.md
drwxrwsr-x.  8 junaid c-pionlt   521 Nov 22 02:48 src
(replay_lt_env) [junaid@ifarm2401 LTSEP_ANALYSIS]$
```

Figure 7: screenshot of `LTSEP_ANALYSIS/src`

The `LTSep_CSVs` directory stores all the CSV files, generated by the analysis scripts, that are used in the L/T/LT/TT separation analysis for the required physics settings. The shell script automatically copies the `LTSep_CSVs` directory from the `UTIL_PION` repository to `LTSEP_ANALYSIS`, because it contains all bin-by-bin normalized physics data yields calculated using the `UTIL_PION` analysis scripts. All the bin-by-bin SIMC yields will also be stored in this directory, and will be calculated using scripts located in the `src` directory. The `iterations` directory will back up all the LT-separation cross-section results, including PDF plots and CSV files for each iteration generated by the analysis scripts. All analysis scripts required to calculate the binned SIMC yields and cross-sections iterations are located in the **src** directory. The `pion_prodYield_ltsep_analysis.sh` shell script has seven flags, and each flag calls a separate Python script in the `src` subdirectory, depending on the analysis requirements. All the required paths, file-name constructions, and Python script arguments are declared in this shell script. The details of the flags are explained in the Table 2. The script requires three arguments. To run this script, execute the following command:

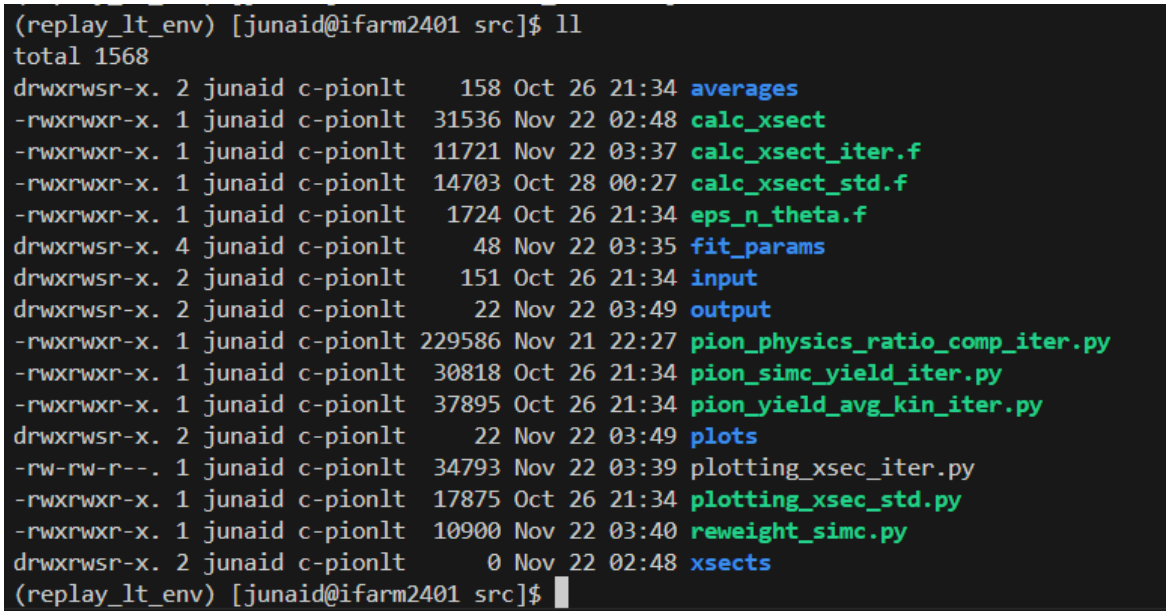
```
./pion_prodYield_ltsep_analysis.sh -flag <Iteration Number> <RunList>
```

The subdirectory `src` contains 6 main Python, 3 Fortran scripts and few more subdirectories (Figure 8) required by the **pion_prodYield_ltsep_analysis.sh** shell script. The **pion_simc_yield_iter.py** script reads the diamond-cut and t-binning information from the CSV files located in the `LTSep_CSVs` directory and calculates the SIMC yields for each t- and ϕ-bin for the empirical SIMC physics model. The output from this script will be written to the `LT_Sep` directory. The **pion_physics_ratio_comp_iter.py** script calculates the DATA/SIMC yield ratios and produces comparison plots for the empirical SIMC physics model. The **pion_yield_avg_kin_iter.py** script computes the average kinematics for each bin and stores them in the required format for the subsequent unseparated cross-section calculation by the Fortran scripts and for the L/T/LT/TT separation analysis. The output from this script is saved in five CSV files with the format explained in Table 3.

The total differential cross-section calculation scripts are written in Fortran language. The `calc_xsect_std.f` script calculates the differential cross-section for the default

Flag	Function	RunList
-s	runs SIMC yield script to determine bin-by-bin SIMC yields	Setting by setting
-r	runs plotting script to compare DATA/SIMC and determine ratios for the empirical SIMC physics model	Setting by setting
-a	runs script to calculate average kinematics and yields for LTSep analysis for the iterative SIMC physics model	Whole Setting
-x	runs script to determine both separated and unseparated cross-sections for the default SIMC physics model (`physics_pion.f`	Whole Setting
-i	runs script to determine SIMC iteration-model unseparated cross-sections	Whole Setting
-p	runs script to determine SIMC iteration-model separated cross-sections and finds new optimized parameters by fitting functional forms to separated cross-sections	Whole Setting
-w	runs to recalculate the new SIMC cross-sections and weights using newly optimized parameters and generate raw SIMC ROOTfile	Setting by setting

Table 2: Description of the flags used in `pion_prodYield_ltsep_analysis.sh`. Note: "setting by setting" means running separately for each left, right, and center setting for a given Q^2 and W. "Whole setting" refers to the complete Q^2, W kinematic setting (no need to specify left, center, and right) and should be run only once.



```
(replay_lt_env) [junaid@ifarm2401 src]$ ll
total 1568
drwxrwsr-x. 2 junaid c-pionlt    158 Oct 26 21:34 averages
-rwxrwxr-x. 1 junaid c-pionlt  31536 Nov 22 02:48 calc_xsect
-rwxrwxr-x. 1 junaid c-pionlt  11721 Nov 22 03:37 calc_xsect_iter.f
-rwxrwxr-x. 1 junaid c-pionlt  14703 Oct 28 00:27 calc_xsect_std.f
-rwxrwxr-x. 1 junaid c-pionlt   1724 Oct 26 21:34 eps_n_theta.f
drwxrwsr-x. 4 junaid c-pionlt     48 Nov 22 03:35 fit_params
drwxrwsr-x. 2 junaid c-pionlt    151 Oct 26 21:34 input
drwxrwsr-x. 2 junaid c-pionlt     22 Nov 22 03:49 output
-rwxrwxr-x. 1 junaid c-pionlt 229586 Nov 21 22:27 pion_physics_ratio_comp_iter.py
-rwxrwxr-x. 1 junaid c-pionlt  30818 Oct 26 21:34 pion_simc_yield_iter.py
-rwxrwxr-x. 1 junaid c-pionlt  37895 Oct 26 21:34 pion_yield_avg_kin_iter.py
drwxrwsr-x. 2 junaid c-pionlt     22 Nov 22 03:49 plots
-rw-rw-r--. 1 junaid c-pionlt  34793 Nov 22 03:39 plotting_xsec_iter.py
-rwxrwxr-x. 1 junaid c-pionlt  17875 Oct 26 21:34 plotting_xsec_std.py
-rwxrwxr-x. 1 junaid c-pionlt  10900 Nov 22 03:40 reweight_simc.py
drwxrwsr-x. 2 junaid c-pionlt      0 Nov 22 02:48 xsects
(replay_lt_env) [junaid@ifarm2401 src]$
```

Figure 8: screenshot of `LTSEP_ANALYSIS/src`

SIMC physics model, while the `calc_xsect_iter.f` script calculates the differential cross-section for the empirical SIMC physics model for each iteration. To execute these Fortran scripts, they must first be compiled by running the following command:

```
gfortran calc_xsect_iter.f -o calc_xsect
```

and then executed as

```
./calc_xsect <iteration> <Physics setting>
```

The shell script automatically compiles the required Fortran script for the differential cross-section calculation, depending on the flag used in the command line. For the `-x` flag called for the `pion_prodYield_ltsep_analysis.sh` shell script, the literal name `std` is used for the `ITERATION` argument, whereas if the `-i` flag is used, the `ITERATION` argument is a two-digit iteration number (e.g. `01`, `02`). The shell script automatically copies the required input files (3) from the `LTSep_CSVs` directory to the `averages` and `input` directories for the Fortran differential cross-section calculation scripts. The input parameter file for the differential cross-section model is located at the path:`LTSEP_ANALYSIS/src/fit_params`. The **plotting_xsec_iter.py** script has two main sections. In the first section, it applies a Rosenbluth fit to the differential cross-sections (coming from the `calc_xsect_iter.f` script) and extracts the L/T/LT/TT separated cross-sections for each t-bin. In the second section, it uses ROOTfit method to fit the σ_L, σ_T, σ_{LT}, and σ_{TT} functional forms to the extracted, separated cross-sections for each t-bin and provides the new optimized parameters for the next iteration. The `pion_prodYield_ltsep_analysis.sh` shell script automatically backs up all the output from the current iteration to the path `LTSEP_ANALYSIS/iterations` and copies the new parameters to the `fit_params` directory for next iteration. The **reweight_simc.py** script uses the new parameters to calculate updated total differential cross-sections and new event-by-event weights using the previous SIMC ROOT file. It then outputs a new SIMC ROOT file with updated event-by-event cross-sections and weights, ready for the next iteration, so that SIMC does not

File name	**Variables**	**Details**
`t_bin_interval`	q2 bin, t bin, phi_bin	Q^2, number of t-bins, number of ϕ-bins, t-bin edges
`list_setting_pion`	ipol, q2, eps, th_pq, tmn, tmx, nbin, kset	Polarity, Q^2, epsilon, central angle, t_{min}, t_{max}, number of bins, setting number
`Eb_pion.dat`	Eb, q2, eps	Beam energy, Q^2, epsilon
`aver_q2value_epsvalue.dat`	r, dr, t-bin number, ϕ-bin number	Ratios, ratio errors, t-bin number, ϕ-bin number
`avek_q2value.dat`	q, dq, w, dw, tt, th_pos	Q^2, Q^2 error, W, W error, central θ, setting number

Table 3: Description of the average-kinematics and settings files used in the LTSep iteration analysis.

need to be rerun repeatedly.

Note: The σ_L, σ_T, σ_{LT}, and σ_{TT} functional forms are implemented in three scripts: `reweight_simc.py`, `calc_xsect_iter.f`, and `plotting_xsec_iter.py` for the model iteration process. If the model does not properly describe the data and one wants to update these functional forms, they must be modified consistently in all three scripts. Each script typically produces three types of output files: ROOT files, PDF plots, and a CSV file. The SIMC ROOT files are stored at the path:`/volatile/hallc/c-pionlt/$USER/OUTPUT/Analysis/SIMC` in the corresponding iteration subdirectory. The PDF and CSV files are backed up to path:`/u/group/c-pionlt/USERS/$USER/hallc_replay_lt/LTSEP_ANALYSIS/iterations`.

5 Concluding Remarks

In this report, a complete procedure for performing the L/T/LT/TT separation of exclusive π^+ electro-production cross-sections has been presented, starting from the raw data and ending with iteratively improved, model-dependent separated cross-sections. The method combines carefully calibrated experimental data, a detailed Monte Carlo simulation (SIMC), and a robust, script-driven analysis framework based on the `hallc_replay_lt`, `UTIL_PION`,`LTSEP_ANALYSIS`, and `simc_gfortran` repositories. Together, these tools enable the extraction of bin-by-bin normalized yields, unseparated cross-sections, and the longitudinal, transverse, and interference components needed for a precise Rosenbluth separation.

A key feature of the framework is its iterative treatment of the SIMC physics model. At each iteration, the DATA/SIMC yield ratios are examined, updated cross-sections are computed using Fortran routines, and new functional fits to σ_L, σ_T, σ_{LT}, and σ_{TT} are performed. The resulting optimized parameters are fed back into SIMC via the `reweight_simc.py` script, allowing the model to be refined without regenerating Monte Carlo events. This workflow, summarized in the flowchart on page 3 and the framework diagram on page 6, provides a controlled way to ensure that the simulation gives an accurate description of the data over the full acceptance and kinematic range.

The analysis structure has been designed to be modular, reproducible, and extensible. The use of CSV-based intermediate files, centralized shell scripts with well-defined flags, and clearly documented directory structures makes it straightforward to repeat the analysis for different Q^2–W settings, to test alternative model parameterizations, or to extend the procedure to future data sets. In the longer term, the separated cross-sections obtained with this framework will be used to extract the pion form factor F_π at high Q^2, providing stringent tests of hadronic structure models and improving our understanding of non-perturbative QCD dynamics.